MLOps Lab Block 2 — Dagster + MLflow Pipeline

1. Pipeline Architecture

Five Dagster assets are defined in `dagster_pipeline.py`:

```
raw_dataset ──► preprocessed_data ──► trained_model ──► model_evaluati
     └──────────────► eda_plots
```

Asset	Input	Output
`raw_dataset`	–	metadata DataFrame + image paths
`preprocessed_data`	`raw_dataset`	train/val PyTorch DataLoaders (80/20 plant-disjoint split)
`eda_plots`	`raw_dataset`	target distribution + biomass vs. age figures
`trained_model`	`preprocessed_data`	trained ResNet18 weights + per-epoch metrics
`model_evaluation`	`trained_model`, `preprocessed_data`	val metrics, training curves, parity plot

MLflow tracking is integrated via the official `dagster-mlflow` resource. Each run logs:

- **9 parameters:** `epochs`, `batch_size`, `learning_rate`, `model_name`, `image_size`, `freeze_backbone`, `max_train_batches`, `n_train`, `n_val`
- **6 metrics:** `train_mse`, `val_mse`, `train_mae`, `val_mae` (per epoch), `final_val_mse`, `final_val_mae`
- **Artifacts:** model checkpoint (`model.pt`), training curves, parity plot, metrics file

2. Setup Instructions

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

Place the dataset folder `mlops_biomass_data/` next to `dagster_pipeline.py`, or set the environment variable:

```
export BIOMASS_DATA_DIR=/path/to/mlops_biomass_data
```

Required packages: `dagster`, `dagster-webserver`, `dagster-mlflow`, `mlflow`, `torch`, `torchvision`, `pandas`, `matplotlib`, `Pillow`, `openpyxl`, `scikit-learn`

3. Running the Pipeline

Dagster UI (recommended):

```
export DAGSTER_HOME="$(pwd)/.dagster_home"
dagster dev -f dagster_pipeline.py
# Open http://localhost:3000 → Lineage → "Materialize all"
```

Dagster CLI:

```
export DAGSTER_HOME="$(pwd)/.dagster_home"
dagster asset materialize -f dagster_pipeline.py --select '*'
```

Programmatic:

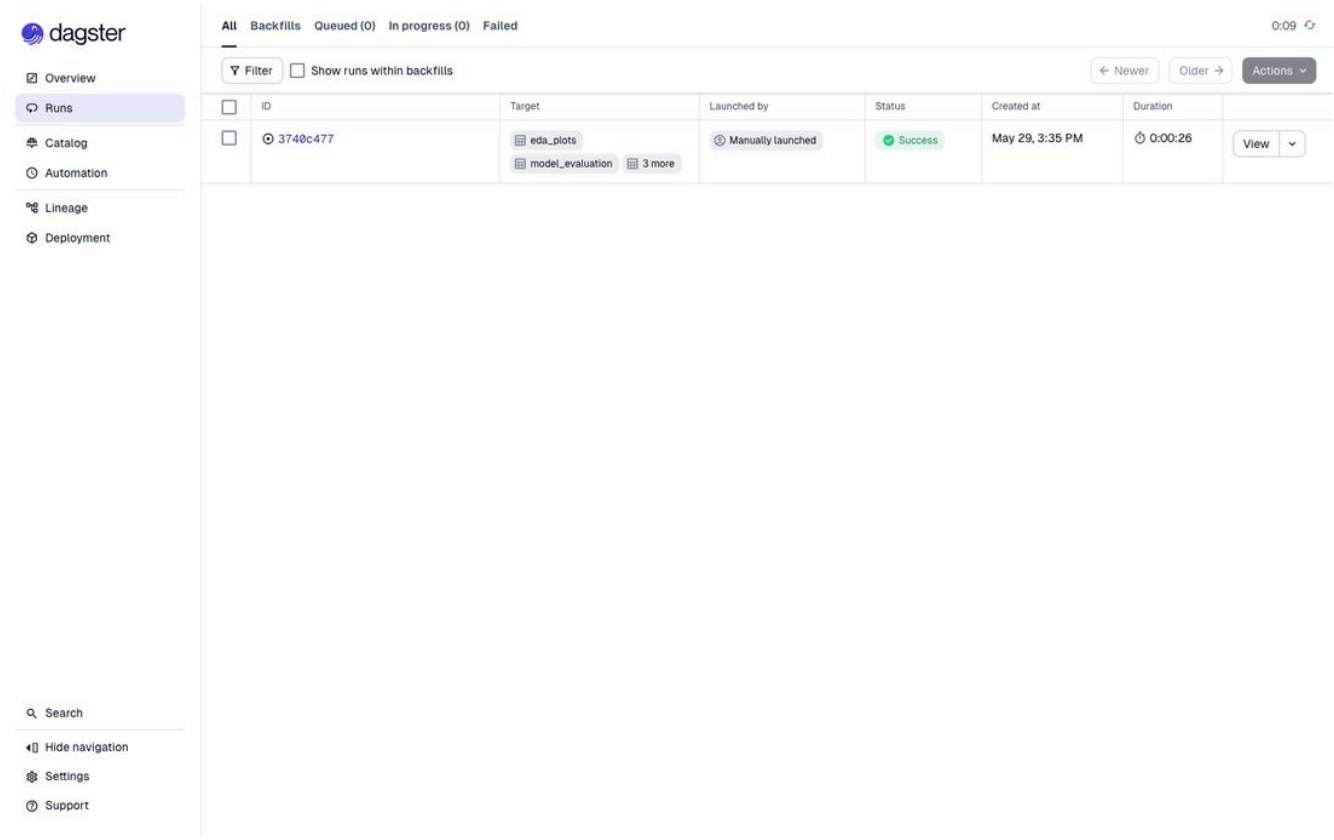
```
python dagster_pipeline.py
```

MLflow UI:

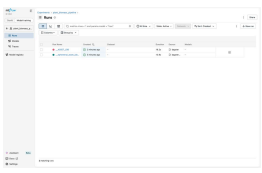
```
mlflow ui --backend-store-uri sqlite:///mlflow.db
# Open http://localhost:5000 → experiment "plant_biomass_pipeline"
```

4. Results

Dagster — successful run



MLflow — experiments



Tracked metrics (2-epoch run, 15 batches/epoch):

Metric	Value
n_train	2,931
n_val	753
Best val MSE	0.0660 g^2
Final val MAE	0.176 g
Target std on val	0.296 g